

Rings: Abelian group (additive) w associative mult. that is distributed in addition

$$(R, \cdot, +)$$

$$\rightarrow \forall a, b, c \in R, \forall a, \exists a^{-1} \text{ s.t. } a + a^{-1} = e$$

$$(a) a + b \in R$$

$$(b) a + a^{-1} = 0 \in R \text{ (additive identity)}$$

$$(c) (a + b) + c = a + (b + c)$$

$$(d) c(a + b) = c \cdot a + c \cdot b$$

$$(e) a(bc) = (ab)c$$

$$(f) \exists 1 \in R \text{ s.t.}$$

$$1 \cdot a = a \cdot 1 = a \quad \forall a \in R$$

$\rightarrow$ All rings are commutative $xy = yx$ UNLESS we are dealing with Matrix groups.

$$* a \cdot 0 = 0$$

Proof: $a + 0 = a \quad \forall a \in R$ but $0 \in R$ so naturally $0 + 0 = 0$. Thus

$$x \cdot (0) = x \cdot (0 + 0) = x \cdot 0 + x \cdot 0$$

• Now, $x \cdot 0 \in R$ and so $\exists$ an additive inverse such that

$$x \cdot 0 + [-x \cdot 0] = 0$$

$$\text{but } x \cdot 0 = x \cdot 0 + x \cdot 0, \text{ from above}$$

So

$$\{x \cdot 0 + x \cdot 0\} + [-x \cdot 0] = 0$$

and so

$$x \cdot 0 + 0 = 0$$

for which we conclude

$$\boxed{x \cdot 0 = 0}$$

* Suppose $1 = 0$, then

$$a \cdot a^{-1} = 1 = 0$$

$$x \cdot 1 = x = x \cdot 0 = 0 \text{ and so } x = 0$$

i.e. $\{R = 0\}$

and the ring is trivial $\forall x \in R$.

Def: A unit is an element u with mult. inverse $\exists \frac{1}{u}$ s.t. $u(\frac{1}{u}) = 1$

$\Rightarrow$ Units form the multiplicative group $R^\times$

ex: $\mathbb{Z}$ has two units $\{-1, 1\}$

i.e. it has a $\mathbb{Z}_2$ unit subgroup

Def: A ring homomorphism is a map between two rings R, R' : $\varphi: R \rightarrow R'$ which preserves sums, products and identities $x, y \in R$, $x', y' \in R'$

$$\varphi(x + y) = \varphi(x) + \varphi(y)$$

$$\varphi(ax) = \varphi(a) \cdot \varphi(x)$$

$$\varphi(x \cdot x^{-1}) = \varphi(1)$$

$$\varphi(x) \cdot \varphi(x^{-1}) = \varphi(1) = 1'$$

$$\varphi(x) \cdot (\varphi(x))^{-1} = 1'$$

* φ is an isomorphism if it is bijjective. i.e. $\exists \varphi^{-1}: R' \rightarrow R$ s.t. $(\varphi \circ \varphi^{-1})(x') = x'$

i.e. it is a linear mapping between rings and can write:

$$\varphi: R \xrightarrow{\sim} R'$$

* φ is an endomorphism if $R = R'$

* φ is an automorphism if $(R = R') \wedge (\varphi \text{ is bijective})$

* $R = R'$ when φ is canonical: if does not depend on any artificial choices, so that R and R' are the same.

$$R[x] = R[y] \text{ when } x \mapsto y$$

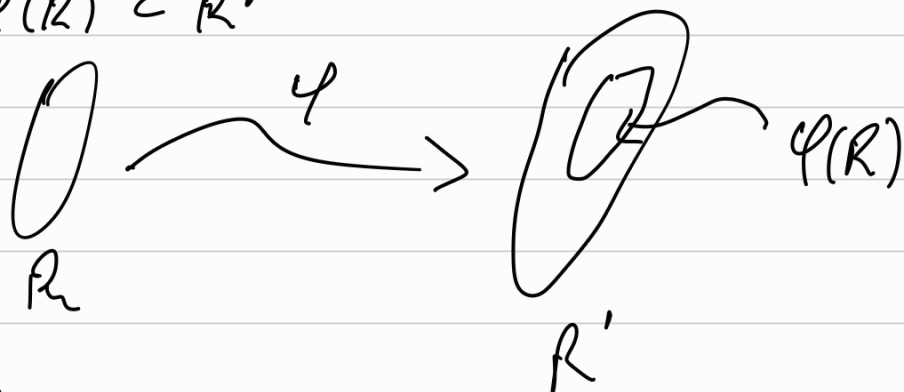
otherwise R and R' are isomorphic and write $R \cong R'$

Def: A subset $R'' \subset R$ is a subring if R'' is a ring and the inclusion $R'' \hookrightarrow R$ is a ring map.

$\Rightarrow R$ is an extension (Ring) of R''

$\Rightarrow R'' \hookrightarrow R$ is an extension (of rings).

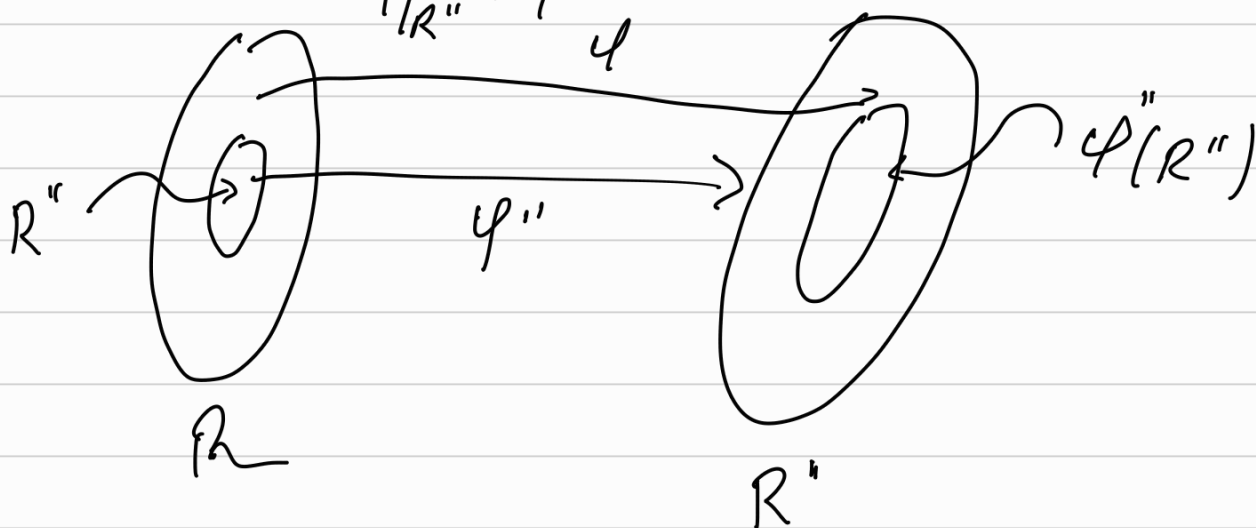
Ex: $\varphi: R \rightarrow R'$, $\text{Im}(\varphi) := \varphi(R)$ is a subset of R'
 $\varphi(R) \subset R'$



$$R'' \subset R \xrightarrow{\varphi} R'$$

* We call $\varphi: R \rightarrow R'$ an extension of $\varphi'': R'' \rightarrow R'$ and we say that φ'' extends to φ if

$$\varphi|_{R''} = \varphi''$$



$$\varphi(R)|_{R''} = \varphi''(R'') \quad \text{on all of } R''$$

* A group G acts on R if $\exists$ a homo given from G into the group of automorphisms of R

$$\text{i.e. } \exists \rho: G \rightarrow \text{Aut}(R)$$

is a homomorphism and in particular, this simply means there exists a representation of G on R .

$\Rightarrow$...

Associate each group element $g \in G$ with its associated automorphism on R

• Let $G \cong GL_n(\mathbb{Q})$, then for $g \in G$

$$p(g): R \rightarrow R$$

and hence acts on R . Often written as

$$g \cdot x = g \cdot x$$

but it should be understood as $p(g)[x] \in R$
 $p_g(x) \in R$.

Def: The ring of invariants R^G is the subring defined by

$$R^G := \{x \in R \mid gx = x \ \forall g \in G\}$$

→ These are all left-invariant elements of the ring. Note L/R inv. are equal since R is commutative.

* G acts on the Quotient R/R' if G acts on R' and each $g \in G$ is an R -map

↳ Back track a bit...

Def: An R -algebra is a ring R' that comes equipped with a ring map $\varphi: R \rightarrow R'$ called a structure-map.

* To indicate that R' is an R -algebra without ref to φ we write R'/R

ex: All rings are canonically $\mathbb{Z}$ -algebras, why?

That would mean $\exists$ a map

$$\varphi: \mathbb{Z} \rightarrow R'$$

$$\forall x_i \in R', \quad \varphi(a) := x_a$$

i.e. we are dealing with "countably infinite" sets. However, can

Q: there not be rings of larger cardinality? or are these only possible for larger structures? DON'T confuse

• An R -algebra homomorphism (R -map) ψ

$$\psi: R' \rightarrow R''$$

which is compatible with their structure maps. $\begin{array}{ccc} R' & \xrightarrow{\psi} & R'' \\ \psi' \uparrow & R & \uparrow \psi'' \end{array}$

$\leadsto$ i.e.

$$\psi(\psi'(x)) = \psi''(x) \quad \text{for some } x \in R$$

Now... A group acts on an R -algebra R'/R $[\psi: R \rightarrow R']$ if G acts on R' and each $g \in G$ is an R -map (into itself)

$$\begin{array}{ccc} R' & \xrightarrow{g} & R' \\ \psi \uparrow & R & \uparrow \psi \end{array}$$

$$g[\psi(x)] = \psi(x)$$

and so R'^G is a R -subalgebra

(Boolean Rings): $\mathbb{F}_2 := \{0, 1\}$

• Given a ring R and a set X let R^X denote the set of functions $f: X \rightarrow R$

Then, R^X is a ring under pointwise addition and mult.

$$R^X := \{f: X \rightarrow R\}$$

$$\leadsto (f+g)(x) = f(x) + g(x)$$

$$(f \cdot g)(x) = f(x)g(x) \quad \leftarrow \text{NOT}$$

Composition

ex: $R := \mathbb{F}_2$ and let $f: X \rightarrow R$, put $S := f^{-1}\{1\} \subset X$

Then, $f(x) = 1$ if $x \in S$ and

$$f(x) = 0 \text{ if } x \notin S$$

• χ_S - Characteristic function χ_S
(Also "step" function)
 $\Rightarrow$ Characteristic functions form a ring

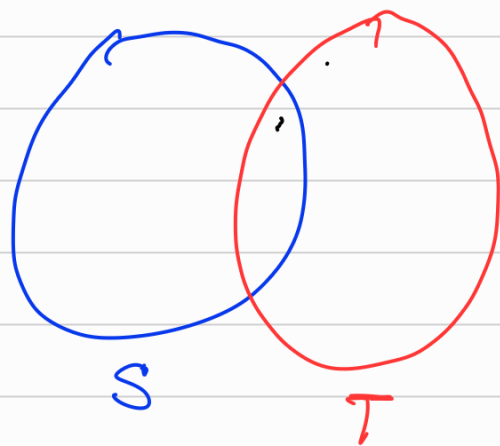
$\mathbb{F}_2^X$
 • Now, take $T \subset X$ then, $x \in X$

$$[\chi_S \circ \chi_T](x) = \begin{cases} 1, & x \in S \cap T \\ 0, & x \notin S \cap T \end{cases}$$

$$[\chi_S \circ \chi_T] = \chi_{S \cap T}$$

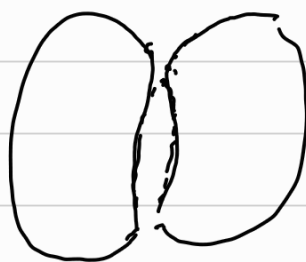
Define the symmetric difference:

$$S \Delta T := (S \cup T) - (S \cap T) = (S - T) \cup (T - S)$$



$\chi_S + \chi_T = \chi_{S \Delta T}$
 • Since $\chi_{S \Delta T}$ cannot be > 1 it must be the disjoint sum of $S \Delta T$

$$S \Delta T =$$



$\Rightarrow$ The subsets of X form a ring:
 (i) Sum is a symmetric difference
 (ii) Product of intersections
 • Canonically isomorphic to $\mathbb{F}_2^X$

(i) Sum is associative

$$S \Delta (T \Delta U) = S \Delta [T \cup U - T \cap U]$$

$$= S \cup T \cup U - S \Delta [T \cup U]$$

$$= S \cup [T \cap U] + S \cap [T \cap U]$$

$$(S \cap [T \cap U]) = S \cap [T \cup U]$$

$$SUTUV - SUTUV \quad \text{⊗}$$

$$\begin{aligned} (SAT) \Delta U &= (SUT - SNT) \Delta U \\ &= SUTUV - SUTNV \\ &\quad - SNTUV + SNTNV \end{aligned}$$

Both the same since $TNU = UNT$ and similarly for other set operands.

- If we break X into functions f_{S_i} s.t. $\bigcup_i S_i = X$ and

$f_{S_i}(x) = 1$, if $x \in S_i$ then the isomorphism $f_{S_i} \cong S_i$ where the ring structure is defined above.

* A ring is called Boolean if $f^2 = f \quad \forall f \in B$.

Note $f^2 = f \Rightarrow f + f = 2f$

Now, $f + f \in B$ and hence must also satisfy $(f + f) = (f + f)^2$, so

$$2f = 4f^2 = 4f \Leftrightarrow 2f = 4f$$

$[2f] = 2 \cdot [2f]$ and hence we conclude

$$\boxed{2f = 0}$$

ex: $\mathbb{F}_2 = \{0, 1\}$

$$2 \cdot 1 = 0 \quad \text{and} \quad 2 \cdot 0 = 0$$

$$1 + 1 = 0$$

Which is how we obtain a $\mathbb{Z}_2$ alg.

* Suppose X is a topological space and give $\mathbb{P}_2$ the discrete topology: Every subset is open and closed.

$f \in C(X, \mathbb{F}_2) \cong X_S$
 where S is both open and closed.

$$f: X \rightarrow \mathbb{F}_2$$

* In order that f be continuous it must be that all open sets in X are mapped to open sets in $\mathbb{F}_2$. However since open sets in $\mathbb{F}_2$ are also closed thus so too are all in X .

→ Form a Boolean subring of $\mathbb{F}_2^X$
Stones Theorem: Every Boolean ring is canonically isomorphic to the ring of cont. functions from a compact Hausdorff top. space X to $\mathbb{F}_2$, or, isomorphic to the ring of cl-open subsets of X .

Polynomial rings:

[Formal Polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

* The set of formal polynomials with coefficients in a ring R is denoted

$$R[x]$$

ex: $\mathbb{Z}[x]$ denotes the set of polynomials with integer coefficients

• Monomials x^i are independent, so

$$g(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0$$

* A monomial is a formal product of n -variables
 $x_1^{i_1} x_2^{i_2} \dots x_n^{i_n}$

with degree $i_1 + \dots + i_n$

multi-index $i = (i_1, \dots, i_n)$

$$x^i = x_1^{i_1} x_2^{i_2} \dots x_n^{i_n}$$

Given a polynomial

$$f(x) = f(x_1, \dots, x_n)$$

we can write the formal expansion

$$f(x) = \sum_i a_i x^i = \sum_{n=1} a_{(i_1, \dots, i_n)} x_1^{i_1} \dots x_n^{i_n}$$

$$* f(x) = a_0 + a_1 x_1^2 + 2a_{12} x_1^1 x_2^1 + a_{11} x_1^2 + a_{22} x_2^2 + \dots$$

where

$$a_i = a_{(i_1, \dots, i_n)} \text{ etc. } \dots$$

* Let R be a ring, $P := R[x_1, \dots, x_n]$ the polynomial ring in n -variables.

Universal Mapping Property: (UMP)

given any ring map $\varphi: R \rightarrow R'$ and an element $x_i \in R'$ for each i , $\exists!$ a ring map

$$\pi: P \rightarrow R' \text{ s.t.}$$

$$\pi|_R = \varphi$$

$$\text{and } \pi(x_i) = x_i.$$

$$\underline{x}_i \in R \text{ and } x_i \in R$$

• Claim:

$$R \xrightarrow{\varphi} R'$$

$$\begin{array}{ccc} & & \nearrow \pi \\ & \searrow & \\ & P & \end{array}$$

• Claim: $R \subset P$ but $P \not\subset R$

$\hookrightarrow$ Take $\alpha \in R$ then if $\alpha = \underline{x}_i$ for some i this is trivial. Now, suppose $\alpha \neq \underline{x}_i$ for all $i = 1, \dots, n$. Then, under closure of R , $\exists \beta, \gamma \in R$ s.t. $\beta + \gamma = \alpha$

However, by the same logic, for γ and β we can have $\gamma = a_1 + a_2 \dots$
 so where we then have a decomp
 $\alpha = \sum_i a_i$
 and this is trivially part of $P[X]$
 hence $\alpha \in P[X]$. However it is
 easy to see that
 we $P[X]$ can exist s.t. $w \notin R$
 since $\sum_i$ can all be s.t. $\sum_i \in R$
 and in fact
 $P[X] \cong R$.

π is given by the formula:

$$\pi \left(\sum q_{(i_1, \dots, i_n)} X_1^{i_1} X_2^{i_2} \dots X_n^{i_n} \right) = \sum \varphi(q_{(i_1, \dots, i_n)}) X_1^{i_1} \dots X_n^{i_n}$$

$$\pi: P \rightarrow R', \quad \varphi: R \rightarrow R'$$

Q: Is this a sufficient condition to make sense of the String-Theory identifications
 $\Omega_{i_1, \dots, i_n} X_1^{i_1} \dots X_n^{i_n} \rightarrow \alpha_{i_1, \dots, i_n} \omega_1^{i_1} \dots \omega_n^{i_n} ??$

T: where X_i are fermionic (super-sym) Quantum fields of some theory on a manifold M and $\omega_i \in \Omega^2(M)$ external forms over the same manifold.

In particular, in Topological string theory when we identify the zero-modes of the A/B-models we obtain the above identifications.

• At the level of co-homologred alg. we have:
 $H^0(P/M) \cong \bigoplus_{i=1}^n H^0(P/M) \otimes \Lambda^i T^{1,0} M$

$$M(\mathbb{Q}_B) = \overline{P} \neq 0 \quad \text{if } (M, 1 \setminus 1 \setminus M)$$

$$Q_B \mapsto \bar{Q}, \quad M^i \mapsto d\bar{z}^i$$

$$\theta_i \mapsto \partial/\partial z^i$$

In expanded form:

$$\pi(\sum a_{i_1 \dots i_n} x_1^{i_1} \dots x_n^{i_n}) = \sum \pi(a_{i_1 \dots i_n}) \pi(x_1^{i_1}) \pi(x_2^{i_2}) \dots \pi(x_n^{i_n})$$

Since π is a homomorphism and $\pi(x_i) = x_i$:

$$\pi(x_{i_k}^{i_k}) = \pi(x_{i_k})^{i_k} = x_{i_k}^{i_k}$$

and $\pi|_R = \varphi$ and $a_{i_1 \dots i_n} \in R$ we conclude the above.

* P is universal among R -algebras equipped with a list of n -elements:

P is one, and P maps uniquely to any other w/ the lists are repeated.

i.e. R/P

• Let $X := \{x_\lambda\}_{\lambda \in \Lambda}$ be any set of variables.

Set $P' := R[X]$

$\hookrightarrow$ polynomials in any finitely many of the x_λ . Sum & product as in P .

* P' is basically the "Power Set" equivalent of polynomials over X . The UMP also applies to P' :

Given $\varphi: R \rightarrow R'$ and $x_\lambda \in R'$ for each λ
 $\exists! \pi: P' \rightarrow R'$ w/ $\pi|_R = \varphi$ and $\pi(x_\lambda) = x_\lambda$.

Ideals:

Let R be a ring. A subset $\triangleleft \subset R$ is an ideal if

(i) $0 \in \triangleleft$ (zero element)

(ii) $a, b \in \triangleleft$, then $a+b \in \triangleleft$ (closure)

(iii) $\forall x \in R, a \in \triangleleft, x \cdot a \in \triangleleft$

* Given a subset $A \subset R$, the ideal $\langle A \rangle$ is the smallest ideal containing A

→ The free group F_n with generators $\{x_1, \dots, x_n\}$ we mean the set of all possible "words" in the "alphabet" $x_1^{-1}, x_1, \dots, x_n^{-1}, x_n$ which can be made.

Def: a group G is finitely generated by $g_1, \dots, g_n \in G$, written as $G = \langle g_1, \dots, g_n \rangle$ i.e. all $g \in G$ can be written as an element of $P[g_1, \dots, g_n]$.

* From this, given any $a_\lambda \in R$ and for $\lambda \in \Lambda$, by the ideal they generate we mean the ideal generated by the set $\{a_\lambda\}_{\lambda \in \Lambda}$

(a) $\Lambda = \emptyset$, $\langle A \rangle = \{0\}$

(b) $\Lambda = \{1, \dots, n\}$, then the ideal is given by $\langle a_1, \dots, a_n \rangle \subset P[a_1, \dots, a_n]$

- Any ideal containing all a_λ contains any (finite) linear combination $\sum x_\lambda a_\lambda$, $x_\lambda \in R$ and at most all 0.

→ By "smallest" the idea is the smallest $I \subset R$ s.t. any linear combination of $I \cdot A$ spans all of A .

→ Any finite group is finitely generated.

a_λ : generators

Given any a we say $\langle a \rangle$ is principal and is all multiples of a
 $\langle a \rangle = Ra$

Proof $\langle a \rangle$ is an ideal:

(a) $0 \in \langle a \rangle$ for $0 \cdot a$

(b) Let $b, c \in \langle a \rangle$ then

$b = na, c = ma$ where $n, m \in R$

then $b + c = (n+m)a = \gamma a \in \langle a \rangle$

and $n+m = \gamma \in R$ so $\langle a \rangle$ is closed under add.

(c) Trivially satisfied.

Sum of ideals: Given ideals

$\langle a_\lambda \rangle$, $\langle a_\sigma \rangle$ where $\lambda \in \Lambda$, $\sigma \in \Sigma$

then sum

$$\sum_{\lambda} \Delta_{\lambda} \equiv \sum_{a_{\lambda} \in \Delta_{\lambda}} x_{\lambda} a_{\lambda} \mid x_{\lambda} \in R$$

* $\sum_{\lambda} \Delta_{\lambda}$ is equal to the ideal that a_{λ} generate

Intersection: $\bigcap \Delta_{\lambda}$ intersection as sets

• If Λ is finite, the Product of R

Def: Given two ideals A and B , the transporter of B to A is the set

$$(A : B) := \{x \in R \mid xB \subset A\}$$

Claim $(A : B)$ is an ideal:

Proof: i.e. left action on B yields a

(i) by def both B and $A \ni 0$
so $x \cdot 0 = 0 \in A \quad \forall$

(ii) let x_1 and $x_2 \in (A : B)$

then for $\forall b \in B \exists a_i \in A$ s.t.

$$x_1 b = a_1$$

$$x_2 b = a_2$$

$$\text{now, } (x_1 + x_2)b = x_1 b + x_2 b = a_1 + a_2$$

but since A is an ideal $a_1 + a_2 \in A$

hence $+$ is closed

(iii) Let $w \in R$ and $x \in (a:b)$ then
must show $wx \in (a:b)$.

$$wx \cdot b = wa$$

now since Δ is an ideal $wa \in \Delta$

In Set notation

$$\Delta \subseteq C \cap D \subseteq \Delta + D$$

$$\Delta, D \subseteq \Delta + D$$

$$\Delta \subseteq (a:b)$$

Claim $\Delta = R$ iff $1 \in \Delta$

Proof: ($\Leftarrow$) $1 \in \Delta$ then for $x \in R$

$$x = x \cdot 1 \text{ but by (iii) } x \in \Delta$$

($\Rightarrow$) Suppose $\Delta = R$ - trivial

If $\langle x \rangle = R$, then x is a unit since

$$\exists y \text{ s.t. } xy = 1$$

* If $\Delta \neq R$, then Δ is said to be
proper

• Given a ring map $\varphi: R \rightarrow R'$, denote
 $\Delta R'$ or Δ^e the ideal of R' generated
by the set $\varphi(\Delta)$. It is the extension
of Δ .

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & R' \\ \cup & & \cup \\ \Delta & \xrightarrow{\varphi} & \Delta R' \end{array} \quad \varphi: \text{ is an isomorphism of } \text{ideals}$$

$\Rightarrow$ proof:

$$(i) \varphi(0) = 0' \text{ and so } \varphi(\Delta) \ni 0'$$

$$(ii) \varphi(a+b) = \varphi(a) + \varphi(b)$$

but since Δ is an ideal

$$a+b \in \Delta \text{ or } a+b = c \in \Delta$$

$$\text{thus } \varphi(c) = \varphi(a) + \varphi(b) \in \varphi(\Delta)$$

$$\text{(iii)} \quad x \cdot a \in \Delta$$

$$\varphi(x \cdot a) = \varphi(x) \varphi(a)$$

$$\varphi(x) \in R' \rightarrow x'$$

$$x' \cdot \varphi(a)$$

$$\text{since } x \cdot a \in \Delta \Rightarrow x \cdot a = \gamma$$

$$\varphi(\gamma) = x' \cdot \varphi(a)$$

so closed

Residue Rings: Let $\varphi: R \rightarrow R'$ be a ring map.
The kernel $\text{Ker}(\varphi)$ of φ is the set
 $\varphi^{-1}(0) = \text{Ker}(\varphi)$

claim: $\varphi^{-1}(0)$ is an ideal of R

proof: (i) $\varphi(0) = 0' \Leftrightarrow \varphi^{-1}(0) = 0_R$

(ii) Let $a, b \in \text{Ker}(\varphi)$, then

$$\varphi(a) = 0' \text{ and } \varphi(b) = 0'$$

$$\text{now, } \varphi(a+b) = \varphi(a) + \varphi(b)$$

$$= 0 + 0$$

$$= 0$$

$$\text{thus } a+b \in \text{Ker}(\varphi)$$

(iii) Let $x \in R$ and $a \in \text{Ker}(\varphi)$

then

$$\varphi(x \cdot a) = \varphi(x) \cdot 0' = 0'$$

$$\text{thus } x \cdot a \in \text{Ker}(\varphi) \quad \forall x \in R$$

* $\text{Ker}(\varphi) = \{0\}$ iff φ is injective.

proof: ($\Rightarrow$) $\varphi(0) = 0$ trivial and
is injective

($\Leftarrow$) Suppose φ is injective. then

$$\varphi(x) = 0$$

$$\varphi(a) = 0$$

$\Rightarrow$ But this must mean $x=0$.

* Let Δ be an ideal of R :

$$R/\Delta := \{x + \Delta \mid x \in R\}$$

(Quotient Ring)

(Set of Cosets)

an equivalence class of elements

$$x \sim y \Rightarrow \exists a \in \Delta \text{ s.t.}$$

$$x - a = y$$

$$R/\Delta := \{[x] \mid x = y - a, a \in \Delta, x, y \in R\}$$

• Also inherits a ring structure

* Called Residue ring, quotient ring, factor ring, R modulo Δ .

• Define the Quotient map: $K: R \rightarrow R/\Delta$
 $K(x) := [x]$

$$[x] = \{y \in R \mid y = x - a\}$$

$[x]$: Called the Residue of x

• K is surjective (trivially)

• Claim: K is a ring map and $\text{Ker}(K) = \Delta$

Proof: (i) $K(x+y)$

$$\bullet K(x+y) = [x+y]$$

$$w = x+y - a$$

$$\text{Since } a \in \Delta \exists \gamma + \delta \text{ s.t.}$$

$$a = \gamma + \delta, \text{ thus}$$

$$w = x - \delta + y - \gamma$$

$$\in [x] + [y]$$

$$\bullet K(xw) = [xw] \Rightarrow \exists a \in \Delta \text{ s.t.}$$

$$\gamma = xw - a$$

now $\forall x \in R$ and $b \in \Delta$

$$x \cdot b \in \mathcal{A}$$

$$a = x \cdot b$$

Suppose $\nexists b \in \mathcal{A}$ s.t. $a = x \cdot b$
 $a + b \in \mathcal{A}$

now, $c = a + b$, by (iii)

$$xc \in \mathcal{A}$$

$$xa + xb = \delta + xb = \gamma \in \mathcal{A}$$

now, the additive inverse of δ ($-\delta$) $\in \mathcal{A}$ by (i) Thus

$$\gamma + (-\delta) = xb \in \mathcal{A}$$

we therefore have a contradiction.

Thus, $\forall a \in \mathcal{A} \exists x \in R, b \in \mathcal{A}$
 s.t. $a = x \cdot b$

$$\gamma = \cancel{xw} - xb = \gamma = x(w - b)$$

$$K(xw) = x[w] \quad \checkmark$$

* K has Kern $\mathcal{A}$.

($\Leftarrow$) Let $x \in \text{Kern}(K)$, then

$$K(x) = [0] \Rightarrow K(x) = 0 + \mathcal{A}$$

$$K(x) \in \mathcal{A}$$

($\Rightarrow$) The zero element in $R/\mathcal{A}$ is

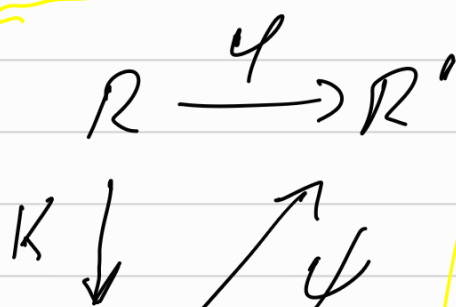
$$[0] \cong \mathcal{A}$$

$$\text{Kern}(K) = \mathcal{A} \quad (\text{proved})$$

* Thus, every ideal is a Kernel of the Canonical Quotient map

Claim: $\text{Kern}(\varphi) \supset \mathcal{A}$ iff $\varphi(\mathcal{A}) = 0$

Proof: $\varphi: R \rightarrow R'$



(Commutative Diagram)

$$R/A$$

($\Rightarrow$) Suppose $A \in \text{Kern}(\varphi)$, then $\varphi(A) = 0$ trivially

($\Leftarrow$) if φ only that's feeling trivial.

* If $A \in \text{Kern}(\varphi)$, then $\exists$ a ring map $\psi: R/A \rightarrow R'$ s.t.

$$\psi \circ \kappa = \varphi$$

Let $x \in R$, then

$$\begin{array}{l|l} \kappa(x) = [x] & \varphi(x) \in R' \\ \psi[\kappa(x)] = \psi([x]) & \end{array}$$

• Note, if $\forall x \in A$, then $\kappa(A) \cong A$

$$\psi[A] = 0' = \varphi(A)$$

$$\bullet \psi([x] + [y]) = \psi([x]) + \psi([y])$$

$$\psi([x]) := \varphi(x)$$



(Defining property)

* Proof of homomorphism:

$$(i) \text{ Claim: } \psi([x] + [y]) = \psi([x]) + \psi([y])$$

we found $[x+y] = [x+y]$ and so

$$\psi([x] + [y]) = \psi([x+y]) = \varphi(x+y)$$

$$= \varphi(x) + \varphi(y)$$

$$= \psi([x]) + \psi([y]) \quad \checkmark$$

$$(ii) \psi(\omega \circ [x]) = \psi([\omega x]) \text{ as proven above. } \varphi(\omega x) = \varphi(\omega x)$$

$$\approx \varphi(w) \varphi(x) = \varphi(wx) \varphi(1)$$

$$\text{Lick: } [a \cdot a^{-1}] = [1]$$

$$\varphi([a \cdot a^{-1}]) = \varphi(a a^{-1}) = \varphi(1) = 1'$$

$$\therefore \varphi([1]) = 1' \quad \checkmark$$

* Conversely, if φ exists, then $\text{Ker}(\varphi) \supseteq A$
 or $\varphi(a) = 0$ or $\lambda R' = 0$ since $ka = 0$

* If φ exists then φ is unique since k is surj.

~> Proof: [I'll try...] Suppose $\exists \rho: R/A \rightarrow R'$

s.t. $\rho \circ k = \varphi$ then

$$\rho \circ k = \varphi \circ k$$

Let $x \in R$, then

$$k(x) = [x]$$

$$\rho([x]) = \varphi([x])$$

• Not sure how surjectivity applies here -

... oh! Since the above holds on ALL of R/A then $\rho = \varphi$ and so φ is uniquely defined.

* If φ exists, then φ is surjective iff φ is too.

• φ is injective iff $A = \text{Kern}(\varphi)$

Proof:

($\Rightarrow$) Suppose φ is injective then

$\forall [x], [y]$ s.t.

$$\varphi([x]) = \varphi([y]) \Leftrightarrow [x] = [y]$$

Now, that implies

$$\varphi(x) = \varphi(y)$$

$$\varphi(w) = 0'$$

$$k(w) = [w]$$

$$\psi([w]) = \psi(w) = 0'$$

But since ψ is injective

$$\psi(A) = \psi([w]) = 0'$$

$$\text{and so } A = [w]$$

$\therefore w \in A$

If ψ is injective then $A = \ker(\psi)$

($\Leftarrow$) Suppose $A = \ker(\psi)$

$$\text{then } \psi(A) = 0'$$

$$K(a) = [a] = A$$

$$\psi(A) = 0'$$

$$\psi([a]) = 0'$$

Suppose ψ was not injective, then

$$\psi([a]) = \psi([b])$$

$$\not\Rightarrow [a] = [b]$$

$$\text{Let } \psi([a]) = \psi([b])$$

$$\psi([a]) - \psi([b]) = \psi([a - b])$$

$$= 0' = \psi([0])$$

$$\psi(a - b) = \psi(0)$$

$$\psi(a) - \psi(b) = 0' \Rightarrow \psi(a) = \psi(b)$$

$$a - b \in \ker(\psi)$$

I see it but im not sure if its safe to just say

$$\psi([a - b]) = 0$$

$$[a - b] \cong A$$

$$K(a - b) = [a - b] = [0]$$

$$\therefore [a] = [b] \quad \checkmark \quad \square \quad ? \dots$$

* ψ is an isomorphism iff ψ is surjective and $A = \ker(\psi)$.

$$\psi: R/\ker(\psi) \xrightarrow{\sim} \text{Im}(\psi)$$

* R/A has UMP: $K(A) = 0$ and given
 $\varphi: R \rightarrow R'$ s.t. $\varphi(A) = 0$, $\exists!$ ring map
 $\psi: R/A \rightarrow R'$ s.t.
 $\psi \circ \kappa = \varphi$

* R/A is universal among R -algebras R'
s.t.

$A R' = 0$
 $\hookrightarrow \begin{cases} a R' = 0, & \varphi: R \rightarrow R' \\ a R' \text{ is the ideal generated} \\ \text{by } \varphi(a) \end{cases}$

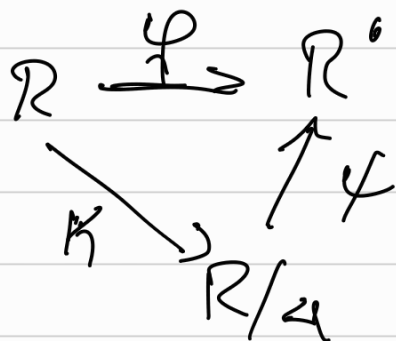
i.e. that the
above is the kernel.

• If A is an ideal generated by a_i , then
the UMP can be rephrased as follows:

UMP alternative form: $K(A_i) = 0 \quad \forall i$, and
 $\left\{ \begin{array}{l} \text{given } \varphi: R \rightarrow R' \text{ s.t.} \\ \varphi(a_i) = 0 \quad \forall i \\ \text{there is a unique ring map } \psi: R/A \rightarrow R' \\ \text{s.t. } \psi \circ \kappa = \varphi \end{array} \right.$
(Proof??)

If A is generated by a_i , then
 $A \subset P[a_1, \dots, a_n]$

now since we have $\varphi: R \rightarrow R'$



UMP states $\exists \pi: P \rightarrow R'$ with $\pi|_R = \varphi$
 and $\pi(\overline{x_i}) = x_i$
 $\rightarrow$ Further, $R/\Delta \subset P[a_1, \dots, a_n]$

$$R/\Delta = \{[x] \mid x + \Delta, x \in R\}$$

Here $\varphi = \pi$ s.t.

$$\varphi(a_i) = 0$$

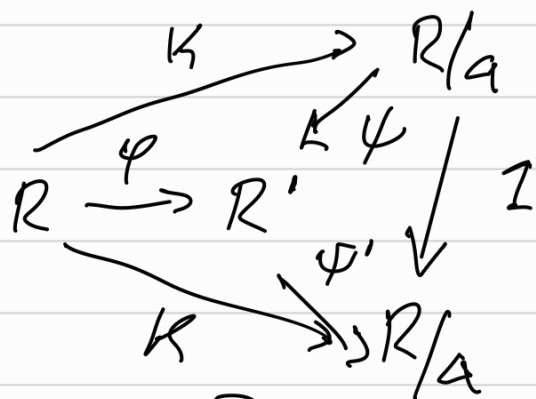
$$\varphi \circ K(\Delta) = \varphi(\Delta) \Leftrightarrow 0 = 0$$

* UMP determines R/Δ up to unique isomorphisms.

$\leadsto$ Say R' equipped w/ $\varphi: R \rightarrow R'$ has the UMP too.

Then $\varphi(\Delta) = 0$ so $\exists! \psi: R/\Delta \rightarrow R'$ with $\psi \circ K = \varphi$ and $K(\Delta) = 0$.

So $\exists! \psi': R' \rightarrow R/\Delta$ with $\psi' \varphi = K$
 $(\psi' \varphi)K = K \Rightarrow \psi' \varphi = \text{id}$



{ UMP: $\pi: P \rightarrow R'$
 $\varphi: R \rightarrow R'$ $P = R[x_1, \dots, x_n]$

• UMP applies to $P = R[x_1, \dots, x_n]$ i.e.
 it applies to R/Δ above.

Proposition: Let R be a ring, $P := R[X]$
 polynomial ring in one variable, $a \in R$ and
 $\pi: P \rightarrow R$ defined by

$$\pi(x) = a$$

Then

$$(i) \text{ Ker}(\pi) = \{F(x) \in P \mid F(a) = 0\} \\ = \langle x - a \rangle$$

$$(ii) P / \langle x - a \rangle \xrightarrow{\sim} R$$

proof:

(i) Define $G := x - a$. Let $F \in P$

• Must show $F = GH + r$, $H \in P$ and $r \in R$. (Not sure why...)

→ I believe there is an isomorphism being used:

$$P[x] \cong K: \{x\} \rightarrow P[x]$$

Let $A \in P[x]$, then we can define this as a function

$$K(x) := \sum a_i x^i$$

and so for each element in $P[x]$ we associate to it its polynomial map. but whatever...

Degree of Polynomial:

Let R be a ring and P the polynomial ring in any # of variables. Let $\forall \neq 0 \in P$ recall the (total) degree $\deg(F)$, is defined as follows:

• If F is a monomial $x_1^{i_1} \dots x_n^{i_n}$ in M then $\deg(M) = i_1 + \dots + i_n$

$$\bullet \deg(F) = \sup(M_i)$$

• Given any $G \in P$ with $FG \neq 0 \in P$, notice

$$\deg(FG) \leq \deg(F) + \deg(G)$$

- Any monomial FG is the product MM of monomial $M \in F$ and $N \in G$
 $\deg(MN) = \deg(M) + \deg(N)$
 $\leq \deg(F) + \deg(G)$

- Case for non-equality:

Suppose $P = P[X]$, and

$$F = aX^m + \dots + a_0, \quad m = \deg(F)$$

$$G = bX^n + \dots + b_0, \quad n = \deg(G)$$

if $ab = 0$, then

$$\deg(FG) < mn$$

- If $a \neq 0$ then $X^2 - (a+b)X$ has $\deg 2$ but has three zeros:
 $\downarrow$ $0, a, b$

Since $ab = 0$

Order of a Polynomial:

Let R be a ring, $P = R[X_\lambda]$ with $\lambda \in \Lambda$ and $(X_\lambda) \in R^\Lambda$ a vector. Let

$$\varphi_{(X_\lambda)}: P \rightarrow P$$

denote the R -alg. map defined by

$$\varphi_{(X_\lambda)}(X_\mu) := X_\mu + x_\mu \quad \forall \mu \in \Lambda$$

- $\varphi_{(X_\lambda)}$ is the automorphism w/ inverse $\varphi_{(-X_\lambda)}$

order of F (at the zero vector (0))

$$\text{ord}_{(0)}(F) := \inf \{ \deg(M_i) \} \text{ of all monomials in } F.$$

order of F at a vector (X_λ)

$$\boxed{\text{ord}_{(X_\lambda)}(F) = \text{ord}_{(0)}(\varphi_{(X_\lambda)} F)}$$

Nested ideals: Let R be a ring, $\mathfrak{a}$ an ideal

and $K: R \rightarrow R/A$ the quotient map.

• Given $b \supset A$, we form the corresp. set of cosets of a

$$b/A := \{b+a \mid b \in b\} = K(b)$$

$\Rightarrow b/A$ is an ideal of R/A

Proof: $[0] \in b/A$, since $0 \in b$, then

$$K(0) = A$$

$$(ii) K(b+b') = K(b) + K(b') \in b$$

(iii) Let $[x] \in R/A$, then

$$[x] \cdot [b]$$

$$(y+a)(b+a') = yb + ab + ya' + aa' \\ = (b+a')y + (ab + aa')$$

Since $a \subset b$ then $a, a' \in b$

$$a = \tilde{b}, \quad a' = b'' \in b$$

$$= (b+b'')y + ab + \tilde{a}$$

$$= \tilde{b} + ab + \tilde{a}$$

$$a \cdot b \in b \quad \text{so}$$

$$\tilde{b} + ab \in b \rightarrow \tilde{b}'$$

$$\therefore = \tilde{b}' + \tilde{a} \in b/A \quad \checkmark$$

$$\text{Also, } b/A = b(R/A)$$

$$\hookrightarrow K(b)$$

$$K: b \rightarrow R/A$$

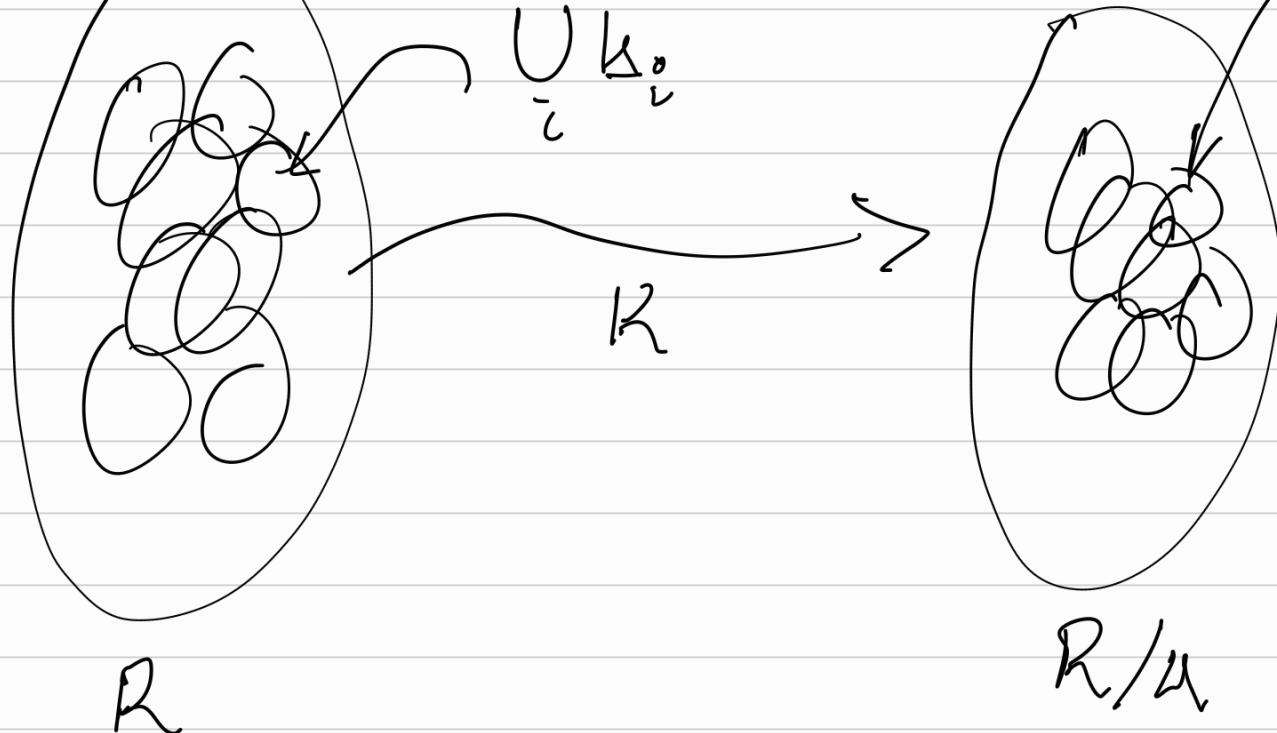
$$* b \rightarrow b/A \quad \text{and}$$

$$b' \mapsto K^{-1}(b)$$

are inverse to each other and establish a bijective correspondence between ideals b of R containing A and the set of all ideals b' of R/A .

$$b' \subset R/A$$

$$\cup b'$$



• This correspondence preserves inclusions
 * Given an ideal $b \supset a$, form the composition of the quotient maps:

$$\varphi: R \rightarrow R/a \rightarrow (R/a)/(b/a)$$

• Since b/a is an ideal of R/a we can form the last set.

$$\begin{array}{ccc} R & \xrightarrow{K^b} & R/b \\ K^a \downarrow & & \varphi \downarrow \cong \\ R/a & \rightarrow & (R/a)/(b/a) \end{array} \quad \text{Ker}(\varphi) = b$$

(Idempotents) Let $e \in R$ be an idempotent i.e. $e^2 = e$. Then Re is a ring with e as 1, because $(xe)e = xe$
 But Re is not a subring of R unless $e = 1$, Re is an ideal of R .

* Set $e' := 1 - e$. Then e' is idempotent

$$e'e' = 1 - 2e - e^2 = 1 - e = e'$$

$$\text{and } e'e = 0$$

$$(1-e)e = e - e^2 = 0$$

~> We call e and e' complementary idempotents

* If $e_1, e_2 \in R$ satisfy

$$\begin{aligned} e_1 + e_2 &= 1 \\ e_1 e_2 &= 0 \end{aligned}$$

then they are complementary idempotents.

For i :

$$e_i = e_i \cdot 1 = e_i (e_1 + e_2) = e_i^2$$

$$i = 1, 2$$

$\text{Idem}(R)$:= Set of all idempotents of R

$\varphi: R \rightarrow R'$, Then $\varphi(e)$ is idempotent:

$$\varphi(e \cdot e) = \varphi(e) \cdot \varphi(e)$$

$$\varphi(e^2) = \varphi(e) \Rightarrow$$

$$\boxed{\varphi(e) = (\varphi(e))^2}$$

$$\varphi|_{\text{Idem}(R)} : \text{Idem}(R) \rightarrow \text{Idem}(R')$$

ex: $R := R' \times R''$

• $(0, 0)$: additive identity

• $(1, 1)$: multiplicative identity

$$e' := (1, 0)$$

$$e'' := (0, 1)$$

Then e' and e'' are complementary idempotents:

$$e' \cdot e' = (1, 0) \cdot (1, 0)$$

$$\equiv (1 \cdot 1, 0) = (1, 0)$$

$$e'' e'' = e''$$

$$e' + e'' = (1, 0) + (0, 1) = (1, 1) = 1$$

$$e' \cdot e'' = (1, 0) \cdot (0, 1) = (0, 0) = \vec{0}$$

Proposition:

Let R be a ring and e' and e'' complementary idempotents

$$\underline{R' := Re'}, \quad \underline{R'' := Re''}$$

Define:

$$\varphi: R \rightarrow R' \times R'' \quad \text{by}$$

$$\underline{\varphi(x) := (xe', xe'')}$$

Then, φ is a ring isomorphism. Moreover,

$$\underline{R' = R/Re''}, \quad \underline{R'' = R/Re'}$$

Proof: Define the surjection

$$\varphi': R \rightarrow R' = Re'$$

$$\varphi'(x) := xe'$$

Claim: φ' is a ring homomorphism

$$\begin{aligned} \text{(i)} \quad \varphi'(x+y) &= (x+y)e' = xe' + ye' \\ &= \varphi(x) + \varphi(y) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \varphi(xy) &= xye' = xy(e')^2 = xe'ye' \\ &= \varphi(x)\varphi(y) \end{aligned}$$

$$\text{(iii)} \quad \varphi(1) = 1 \cdot e' = e' \quad \leftarrow e' \text{ is } 1 \text{ of } R'$$

$$\underline{\text{Claim: } \ker(\varphi') = Re''}$$

$$\text{Proof: } \varphi'(x) = xe' = 0$$

note since $e'e'' = 0$ then

$$x = x \cdot 1 = x(e' + e'') = \underline{xe''}$$

thus,

$$\varphi'(x) = 0 \Rightarrow \underline{x = xe''}$$

$$\underline{x \in Re''}$$

$$\text{Thus } \underline{R/Re'' \cong R'}$$

Similarly, define the surjection

$$\varphi'': R \rightarrow R''$$

$$\varphi''(x) = xe''$$

$$\text{Ker}(\varphi'') = R' = Re'$$

$$R'' = R/Re'$$

φ : is a ring map

$$\begin{aligned} \bullet \varphi(x+y) &= ((x+y)e', (x+y)e'') \\ &= (xe', (x+y)e'') + (ye', 0) \\ &= (xe', xe'') + (0, ye'') + (ye', 0) \\ &= (xe', xe'') + (ye', ye'') \\ &= \varphi(x) + \varphi(y) \end{aligned}$$

$$\begin{aligned} \bullet \varphi(xy) &= (xye', xye'') \\ &= (xe'ye', xe''ye'') \\ &= (xe', xe'') \cdot (ye', ye'') \\ &= \varphi(x)\varphi(y) \end{aligned}$$

$$\bullet \varphi(1) = (e', e'') \quad \checkmark$$

* φ is surjective:

For arbitrary $(xe', x'e'')$ its generator in R is $(xe' + x'e'')$

$$\begin{aligned} \varphi(xe' + x'e'') &= [(xe' + x'e'')e', (xe' + x'e'')e''] \\ &= (xe', x'e'') \end{aligned}$$

* φ is injective:

$$\text{If } \varphi(x) = \varphi(y) \Rightarrow x = y$$

$$\bullet \text{If } xe' = 0 \text{ and } xe'' = 0$$

$$\left(\begin{aligned} x &= x \cdot 1 = x(e' + e'') = 0 \\ \Rightarrow \varphi(x) &= (0, 0) \end{aligned} \right.$$

$$\Rightarrow (xe', xe'') = (ye', ye'')$$

$$((x-y)e', (x-y)e'') = (0, 0)$$

$$\text{so } (x-y)e' = 0$$

$$(x-y)e'' = 0$$

$$\therefore (x-y) = 0 \Rightarrow x = y \quad \checkmark$$

$\Rightarrow$ (φ is bijective) $\Rightarrow \varphi$ is an isomorphism

EXERCISES:

(1.13) Let $\varphi: R \rightarrow R'$ be a map of rings
 $\mathfrak{A}, \mathfrak{A}_1, \mathfrak{A}_2$ ideals of R
 $\mathfrak{B}, \mathfrak{B}_1, \mathfrak{B}_2$ ideals of R'

Prove the following

$$(1a) (\mathfrak{A}_1 + \mathfrak{A}_2)^e = \mathfrak{A}_1^e + \mathfrak{A}_2^e$$

$$\mathfrak{A}_1^e = \varphi(\mathfrak{A}_1)$$

$$\begin{aligned} \varphi(\mathfrak{A}_1 + \mathfrak{A}_2) &= \varphi(\mathfrak{A} + \mathfrak{B}) & a \in \mathfrak{A}_1 \\ &= \varphi(\mathfrak{A}) + \varphi(\mathfrak{B}) & \forall b \in \mathfrak{A}_2 \\ &= \mathfrak{A}_1^e + \mathfrak{A}_2^e \end{aligned}$$

$$(1b) (\mathfrak{B}_1 + \mathfrak{B}_2)^c \supset \mathfrak{B}_1^c + \mathfrak{B}_2^c$$

$$\varphi^{-1}(\mathfrak{A}') = \mathfrak{A}^c$$

$$\mathfrak{B}_1^c + \mathfrak{B}_2^c \subset (\mathfrak{B}_1 + \mathfrak{B}_2)^c$$

$$\varphi^{-1}(\mathfrak{B}_1) + \varphi^{-1}(\mathfrak{B}_2)$$

Let $w \in \mathfrak{B}_1^c + \mathfrak{B}_2^c$, then

$$R \ni w = \varphi^{-1}(\mathfrak{B}_1) + \varphi^{-1}(\mathfrak{B}_2)$$

then let $b_1 \in \mathfrak{B}_1, b_2 \in \mathfrak{B}_2$

$$\varphi(w) = \varphi(b_1) + \varphi(b_2)$$

$$= \varphi(b_1 + b_2)$$

$$\therefore w \in \varphi^{-1}(\mathfrak{B}_1 + \mathfrak{B}_2)$$

$$(2a) (\mathfrak{A}_1 \cap \mathfrak{A}_2)^c \subset \mathfrak{A}_1^c \cap \mathfrak{A}_2^c$$

• Let $\alpha \in \mathfrak{A}_1 \cap \mathfrak{A}_2$

then

$$\varphi(\mathfrak{A}_1 \cap \mathfrak{A}_2) \subset R'$$

If $\mathfrak{A}_1 \cap \mathfrak{A}_2 = \emptyset$, then $\varphi(0) = 0$

Suppose $\mathfrak{A}_1 \cap \mathfrak{A}_2 \neq \emptyset$, then let

$$\alpha \in \mathfrak{A}_1 \cap \mathfrak{A}_2$$

$$\text{Let } x_1 \in \mathfrak{A}_1 \setminus (\mathfrak{A}_1 \cap \mathfrak{A}_2)$$

$$x_2 \in \mathfrak{A}_2 \setminus (\mathfrak{A}_1 \cap \mathfrak{A}_2)$$

then $\exists x_0$ s.t.



$$\alpha_1 \in \mathfrak{A}_1$$

$$\alpha = x_1 \cdot \alpha_2 = \alpha_1 \cdot x_2 \quad \alpha_2 \in \mathcal{A}_2$$

$$\begin{aligned} \text{Thus, } \varphi(\alpha) &= \varphi(x_1 \cdot \alpha_2) = \varphi(\alpha_1 \cdot x_2) \\ &= \varphi(x_1) \cdot \varphi(\alpha_2) = \varphi(\alpha_1) \varphi(x_2) \\ &= \end{aligned}$$

\